

ALTERNATIVE ENERGY SOURCES

Cornellengineer



Going . . .



Illustrations by Jeremy Protska

A pastel dawn awakens you to a gentle April morning. Through your window you see the grass is tender and green, the magnolia trees are budding and the tulips finally breaking through the moist earth. You stretch languidly and tumble out of bed. Eager to feel the wholesome first days of spring, you open your window and draw in a deep refreshing breath of...exhaust? Who is responsible for ru-

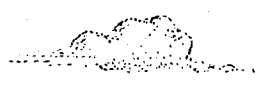
ining your otherwise perfect morning? It's those dastardly, dirty fossil fuels: oil, natural gas, and coal. But why do we use fossil fuels if they pollute our environment so recklessly? Because they are easily accessible, energy rich, and highly portable. Yet because they must be extracted from the earth, they are necessarily in finite supply. All too soon we will have no choice but to abandon fossil fuels in favor of other energy sources, alternatives which we hope are as environmentally sound as they are efficient.

For a moment, try to recall what transported the stench of the automobile exhaust to your nose. Why, it was a breeze blowing across the highway, or maybe it was the chronic gale that whips the crisp winter air into our faces, or the gentle zephyr that keeps our kite aloft. But is this all that wind is good for? Actually, global winds are an environmentally benign, inexhaustible energy source that have yet to be explored.

The use of wind as an energy generating medium has a long and colored history, beginning in Persia some 2,000



... With the Wind



*People from around the world have been
using wind power for millennia.
How can we harness this ancient energy source
for our future energy needs?*

VINCENT ZIMMER

years ago. In that faraway land, windmills harnessed kinetic energy for agriculture and industry, grinding grain for bread and pumping water for domestic use and irrigation. In fact, wind was an energy source second only to the wood burning. Wind ranked with the muscle power of man and his beasts of burden as a primary force by which civilization was created.

In the mid-nineteenth century, the wind supplied one-quarter of the United States' transportation energy, predominantly through the use of large sailed ocean-going ships. In addition, the windmill provided electricity for rural areas and farms of all sizes. The isolated locales and prohibitive cost of running electrical lines to these sites made the windmill an ideal accessory. But with the onset of the Great Depression and Roosevelt's New Deal for an unemployed population, the farms relinquished their windmills as they were linked to utility power grids under the

auspices of the Rural Electrification Administration.

The abundance of cheap fossil fuels, beginning in the 19th-century with largescale coal mining, encouraged the total energy consumption of the US to soar to its now rapacious heights. Current US energy consumption exceeds 250 billion kilowatt hours per year. This unsettling figure is only one example of the exponential growth in energy usage experienced throughout the world. As the earth's population continues to grow geometrically, and as the third world vies for the amenities and power of the first world, such reckless depletion of the earth's resources and the corresponding generation of waste products will increase tenfold. US energy usage in the year 2000 might very well reach an annual figure of 600 billion kilowatts.

Yet the energy woes of our nation in particular, and the world in the general, might be allayed by the employment of wind generated power. In fact,

the Solar Energy Panel of the National Science Foundation has estimated that the winds which now surround the planet have a total power capacity on the order of 100 billion gigawatts. Winds racing across the continental US alone would account for about 100,000 gigawatts of this bounty, a figure which is more than 30 times the amount of energy the nation consumed in 1980.

Harnessing the Wind

Tapping the power of the continental gusts is an extremely difficult and inefficient process. Wind power's great inefficiency involves the physics of the air as well as mechanical aspects of wind turbines themselves. Also, the winds are by nature unreliable and capricious. Today, wind engineers estimate that between 10 and 20 years of wind pattern observations are necessary in order to plan an optimal wind power generation field, since wind patterns vary over



From In the Month of July, Paul Joseph Constantin Gabriël

periods of years.

To build these immense fields of wind power generation, or wind farms as they are called in the industry, the wind turbines are situated in successive rows, or units. The separation between individual units is usually five times the diameter of the propellers (which extend beyond 20 meters for some turbines) in order to avoid having

turbulent air from one turbine entering the path of another turbine and slowing it down. These farms are positioned to optimize the wind which hits each unit. Furthermore, the farms include many units in order to take advantage of economies of scale. The starting capital for a wind farm includes costs for transmission lines from the isolated farm to the power grid, generators

which convert the wind energy to electricity, and storage devices which are often necessary to provide a continuous flow of electricity from a source which is inherently cyclic.

Wind farms have been established most successfully in the American West. The California coast features many narrow mountain passes which funnel powerful, concentrated, and reliable

streams of wind into their respective valleys. Within these valleys, wind farms have been erected, most notably in Tehachapi Pass, Altamont Pass, and San Geronio Pass. Even these efficient, progressive means of providing the nation with a little extra electricity have incurred the wrath of a few Americans. Mayor Sonny Bono of Santa Cruz has alternately been a friend and foe of the wind farms. Most recently upon discovering that if Santa Cruz incorporates wind farms, it could garner additional millions in tax dollars, he began adamant endorsement.

One strike against wind farms arises from the fact that the wind turbines generate an appreciable amount of noise. Houses within one-half mile of a farm might have to contend with noise levels on the order of 50dB, comparable to the drone of a blender or dishwasher. Such background noise is trifling

when juxtaposed with the clatter of a large metropolitan region, but in these quiet rural areas, any disturbances are accentuated. "Once the neighbors are

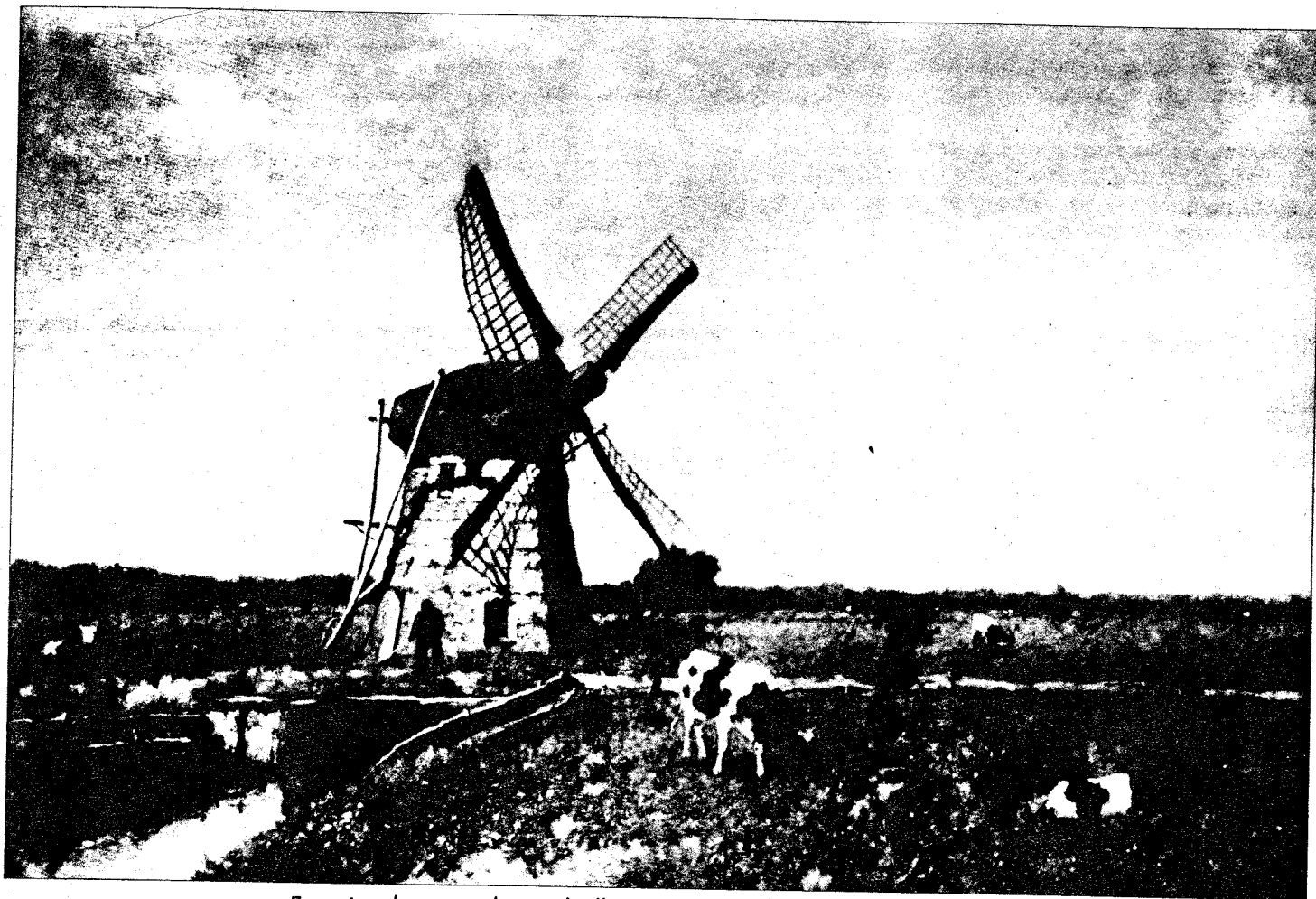
*Winds racing across the
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used to the sound," one disgruntled Norwegian wind farm expert put it, "they will hear it even with a pillow over their ears."

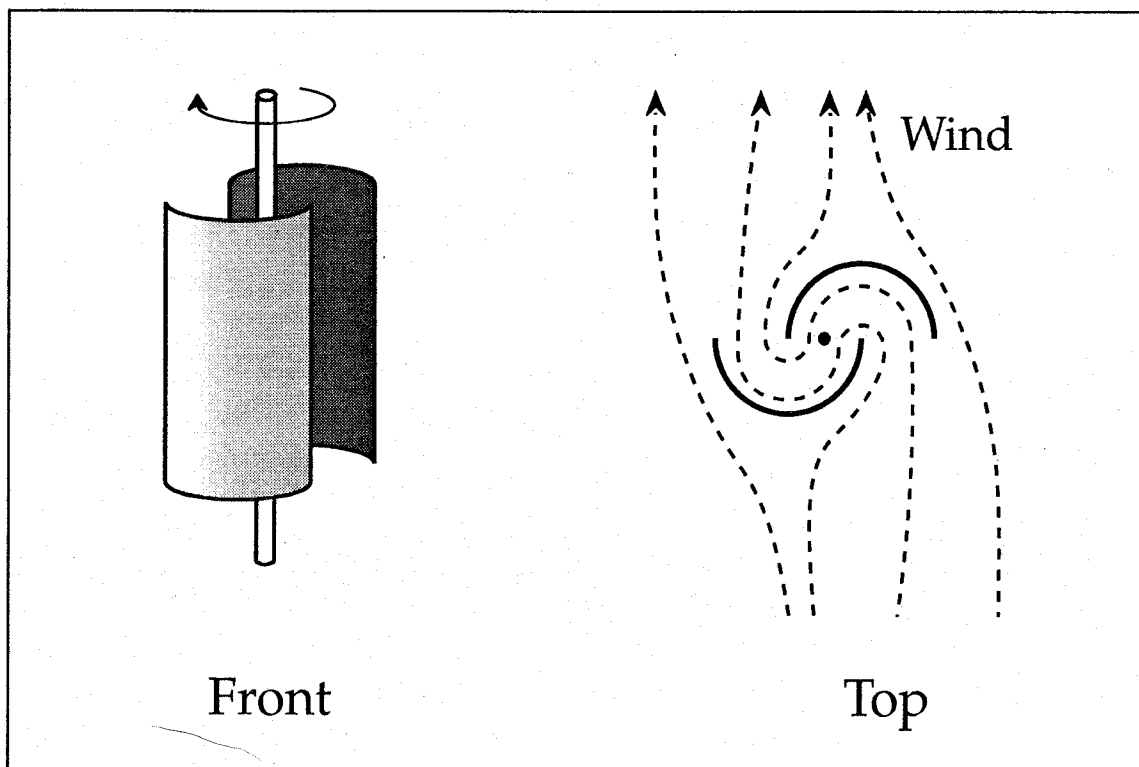
Another point which has impeded the growth of wind farms on the west coast concerns wildlife. Birds have a difficult time navigating the wind farms

and usually cannot see the swiftly moving blades. Consequently, many meet a grisly demise. The Audubon Society and its 55,000 members fear for the fate of the endangered California Condor and the American bald eagle.

Despite the objections to wind driven turbines, wind is one of the forces of nature still stronger than man, and instead of living meekly under her yoke, the engineering community should rise to the challenge of building turbines which can make full use of the phenomenon. The extraction of power from the wind entails many perils, including vibration and structural problems in the turbines. To withstand these, the propellers of wind turbines must be strong enough to endure the stress induced by wind coming at all speeds and directions. If you ever had a pinwheel as a child, recall the deformation of the small



From Landscape with Windmill near Schiedam, J. Hendrik Weissenbruch



The Savonius Rotor and its stream flow

rapidly for the yaw to adjust, the currents will cause different forces to occur at the top and bottom of the blades because the angles of incidence will vary to a large extent. These divergent forces will again cause vibrations in the device. Cornell professor Dennis Shepherd comments: "the vibrations are extremely difficult to model mathematically. We just don't know enough about the conditions met by the turbines." This type of problem is similar to that faced by a helicopter, though, and the manner in which a helicopter can equalize the lift on both of its sides might

Computer Graphic by Joel Millett

be useful study for the designers of wind turbines.

blades even in a light breeze. Now generalize that to a device which is hundreds of times larger. Also, the rotation of a turbine's blades acts as an immense gyroscope, limiting the ability of the rotor to pivot to face the wind. And if this fact were not enough, the problems are compounded by the shear weight of the blades and their housing, which generate torques that push even the strongest materials to their limit.

However, mere physical limitations are not the only problem. The modern wind turbine is a substantial capital investment which must remain functional for at least fifteen years in order to produce any appreciable rate of return. The blades must be of such a constitution that years of varying meteorological conditions do not destroy or debilitate them. Composite fiberglass and epoxy blades would meet these specifications.

Every aspect of wind generated power presents physical problems to be overcome in the structural design of turbines. The first of these barriers involves vibrations within the rotating machinery and blades which occur because of the rapidly changing forces acting at both the blades and the supports. Wind does not exhibit uniform

strength; individual gusts vary in intensity with their height, gaining more speed the farther they fly above the earth. For a turbine's blades, this means substantially stronger forces act on the top of the blade than on the bottom. These forces set the blades vibrating relative to both the axle and each other.

Vibration can be minimized somewhat by using three blades rather than four, but recent attempts to harness more of the air with longer blades has only aggravated the problem. The subsequent vibrations vary the angle between respective propellers and the coning angle of the device. Coning is the case where the blades are not at exact right angles to the shaft but lean slightly downwind in the direction of the winds' force. As a result, the rotating blades do not describe a disc in a plane; instead, they sweep out a shallow cone. Turbulence in the wake of the device, or its "wind shadow," can aggravate the vibration problem if it is not diffused by the leeward end of the device.

The yaw mechanism—the means by which the blades are directed to face the wind—cannot always move quickly enough to keep up with wildly fluctuating gusts. If the winds change too

Notwithstanding the physical constraints with which the wind turbine must constantly comply, the devices of today must also contend with economic reality. The cost of propellor fabrication has increased so much that it now accounts for approximately half of the final cost of the wind turbine. This has led to the adoption of a two-bladed turbine and its minimalist cousin, the one blade with counterweight. The latter can rotate fast enough to maximize wind usage through its propellor circle. This means that breezes of around ten miles per hour are necessary for the blades to rotate in a manner which initiates electricity output. The one-bladed propellor reduces the cost of the generator and other components, components which no longer have to satisfy the stringent mechanical constraints of multi-blade turbines.

Turbine Design

Most wind turbines have been of the traditional, horizontal axis variety (see Figure 1). But since ancient times, windmills have also appeared with their sails or propellers rotating about a ver-

tical axis (see Figure 2). Such vertical windmills were used in Crete to irrigate fields from the days of Homer, while China and Thailand still use this type of sailed windmill to irrigate rice paddies.

The simplest of the modern types, the Savonius rotor (see Figure 3), is fabricated from two half-cylinders. The cylinders are mounted as to face each other in opposite directions, forming an S-shape. The Savonius rotor has an advantage over wind turbines with blades because it can begin supplying power to electric lines at the meager wind speed of five miles per hour, whereas the the conventional horizontal axis wind turbine (HAWT) requires a minimum of ten miles per hour in order to accomplish the same output.

Industrialized nations where capital expenditure dictates that each wind turbine produce power on a large scale, have not smiled upon the Savonius because its S-shape requires too much material, whether it be metal or some composite. In addition, the Savonius' large, cylindrical rotors are vulnerable to storms and other excessive gusts of wind. The rotors cannot pitch and yaw in a storm, and must absorb the full brunt of the gales. The disturbance is then transmitted to the supports, which often break under the strain.

One bright point of the Savonius is that it does operate economically on a small scale; it does not require the large towers of horizontal axis machines. In-

deed, Savonius generators can even be mounted upon houses. In this scenario, the low wind speeds necessary for power generation would enable the Savonius to provide a flow of electricity in most meteorological conditions.

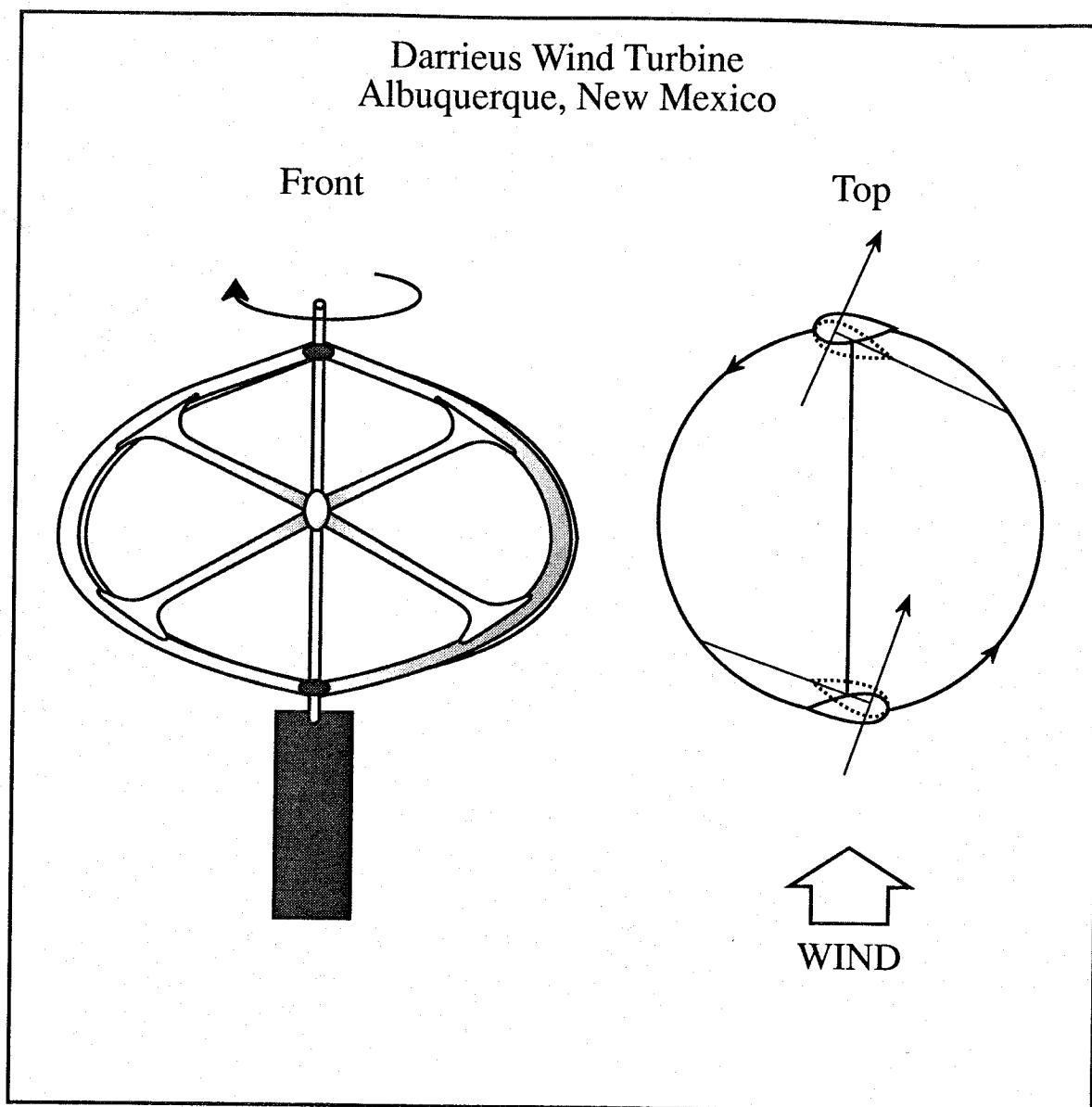
On the other hand, the field is still open for the vertical access wind turbine (VAWT). The small surface area of today's devices, when shaped as an airfoil (i.e., a teardrop with the large portion facing the wind), manage to trap a good deal of the incident wind. The Darrieus Eggbeater (see Figure 4) makes good use of this advantage.

The blades for the Eggbeater are bent in a manner which minimize the stresses of regular operation. The blades

are initially bent and attached to a vertical axis in a manner similar to the household eggbeater. In fact, if a rope is rotated rapidly, the centrifugal force which defines its curvature is the same as that which designers use to model the blades. For a similar picture, imagine the shape of a jump rope when children are playing in the school yard.

This design model implies that ideally, at least, pure tension is the only force which must be countered within the blades. The blades thus require less structural support, and consequently can use propellers lighter than those in the HAWT.

Although the Darrieus is the most successful VAWT designed, it still has



The blades of the Darrieus Turbine pivot to transmit maximum power.

problems. Designing the device for severe storms is problematic, if not impossible. No algorithm yet invented can encompass all the idiosyncrasies of Mother Nature. Supercomputation can provide models which have more parameters and more detail, but the utility of these qualitative simulations has fallen short of expectations. Furthermore, the subtle manner in which the wind drives the Darrieus make it slightly less efficient than that of the propellor-type HAWT.

Ocean Winds

Even though these two devices act as the predominant models for wind turbines, the application of wind turbine technology should not be limited to land. In fact, the placing of wind turbines offshore has been studied by many Scandinavian countries, India, the United States, and Great Britain. An ocean location would be preferable to even the Savannah and Great Plains because the power density of the winds off the coast is ten times that of winds on land—simply because the winds are a result of the uneven heating of the globe by the sun and the rotation of the planet itself. The air remains cooler at sea because the water can absorb more of the sun's energy, whereas on land, the air is warmed, becoming lighter and rising. This temperature gradient (difference between aquatic and terrestrial temperature) causes the cooler sea air to come inland and replace the rising air. Therefore, at the land sea boundary we find a large, consistent movement of air—the sea breeze.

In addition to favorable winds, placing wind turbines offshore has the advantage of locality. The regions immediately off the coasts of the United States and other nations are used neither for habitation nor for agriculture. These large virgin strips around the land masses are only occasionally disturbed by sailing vessels and fisherman. By judiciously situating the wind turbines, the ocean-going traffic and fishing activity could coexist with these permanent structures.

Several objections to offshore wind

turbines concern the safety of winged and aquatic wildlife, navigation difficulties, and the possible unsightliness of these structures. The wind turbines would pose the usual threat to birds, a sad but unavoidable fact. But as for fish, there would be no significant effect beyond perhaps frightening them away with the noise from the devices. And if the turbines were properly illuminated and well-spaced, navigation would not suffer in the least from their presence.

Even with the above considerations neglected, serious impediments still exist regarding the siting of these devices offshore. Recall that the varying winds which the land based turbines faced led to vibration and stress in the devices. Try to imagine those same winds on the scale of gales, and you will see the added trouble faced by designers of offshore wind turbines. The winds off the coast are several orders of magnitude larger than those on land,

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and the foundation of the offshore class, whether it be firm or floating, is not as substantial as the terrestrial variety. Also, the combination of wave and air motion would induce unpredictable vibrations. Moreover, as coastal denizens will testify, the brine in the air wreaks havoc upon any metal surface, hence the components of the wind turbines will be vulnerable to salty moisture which spells inevitable destruction of many turbine components. Finally, it would be expensive and more dangerous to service the offshore models.

Thankfully, these problems have not killed the programs for offshore wind turbines in many countries. The high energy density of the offshore air makes the advantages of the devices far outweigh its drawbacks. The Netherlands has erected a 750 kilowatt wind

turbine offshore and is presently studying the effectiveness of the design. In addition, the United Kingdom has planned to erect such turbines to take advantage of the substantial winds which hit its rocky coasts from the North Sea. Unfortunately, in the United States, red tape has buried the issue in the planning stages.

A Worthy Investment

What will motivate the adoption of the devices now? The economic motive alone should be sufficient; the indirect costs of pollution and import taxes avoided by wind turbines make them a palatable alternative to fossil fuels. More funds must go toward pollution reduction measures every year, an expense which wind power would never produce.

The government has made some progressive gestures toward this issue, such as providing tax incentives for people who successfully develop wind and solar energy generating capabilities. Utilities are now required to buy the excess electricity generated by privately owned windmills. Unfortunately, this incentive is not large enough to encourage much private activity. This lack of activity is happily viewed by utilities who usually pay rock-bottom prices for this electricity and fail to cooperate with the solar or wind farms. The government must step in and encourage the pioneers of renewable energy resources, while at the same time levying a tax on the carbon dioxide emissions by both industry and individuals. In the end, people will only be motivated to change their prodigal ways if their wallets are affected.

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References

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Vincent Zimmer, a senior in EE, has recently been sighted on horseback, charging windmills.